FlowBench: Benchmarking LLM Agent Behavior Against Human Usability Patterns on the Web

Wee Joe Tan*†
joe.tan.25@ucl.ac.uk
University College London
London, United Kingdom

Zi Rui Lucas Lim*†
zi.lim.25@ucl.ac.uk
University College London
London, United Kingdom

Shashank Durgad*†
shashank.durgad.25@ucl.ac.uk
University College London
London, United Kingdom

Karim Obegi*†
karim.obegi.25@ucl.ac.uk
University College London
London, United Kingdom

Aiden Yiliu Li*†‡
yiliu.li.23@ucl.ac.uk
yiliu.li@outlook.com
University College London
London, United Kingdom

Abstract

Whether LLM agents can reliably serve as synthetic users in usability studies rests on an assumption that has yet to be systematically tested: that agent behavior meaningfully mirrors how real humans experience websites. We present **FlowBench**, the first benchmark for measuring behavioral similarity between LLM agents and human users on real-world web tasks. FlowBench comprises 10 diverse websites spanning e-commerce and SaaS domains, each paired with a website-specific task completed by 20 human participants under structured usability protocols. Similarity is evaluated across three dimensions: System Usability Scale (SUS) scores, step-wise Single Ease Question (SEQ) trajectories, and Think Aloud transcripts assessed via semantic embedding cosine similarity. These are aggregated into a unified **FlowBench score**, where 1.0 indicates perfect behavioral alignment. We instantiate the benchmark using OpenFlo as the reference agent harness and report preliminary results, establishing a reproducible foundation for evaluating the fidelity of LLM agents as proxies for human usability testers.

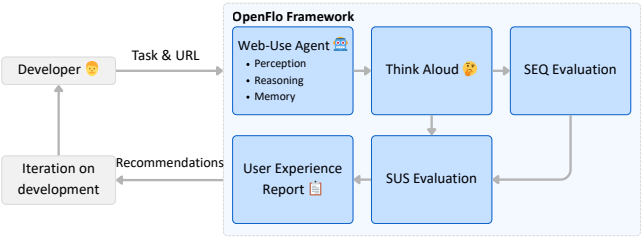

Figure 1: FlowBench measures behavioral similarity between persona-matched human participants and LLM agents across three dimensions, aggregated into a single FlowBench score.

1 Introduction

The growing use of LLM agents as synthetic users in usability research rests on a largely unvalidated assumption: that agent behavior meaningfully reflects how real humans experience and navigate websites. While recent work has demonstrated that agents can automate web tasks [7, 14], the degree to which their evaluative judgments, navigation patterns, and qualitative reasoning align with those of real users remains unmeasured. Without a benchmark grounding these comparisons, claims about agent fidelity to human behavior remain speculative.

Traditional UX evaluation relies on human participants in controlled studies [9, 10], capturing subjective experience through instruments like the System Usability Scale (SUS) [2] and Single Ease Questions (SEQ) [12], alongside qualitative Think Aloud protocols. LLM agents offer the promise of scalable, continuous usability evaluation—but only if their behavioral profiles demonstrably correlate with those of real users.

Existing benchmarks for web agents, such as WebArena [15] and Mind2Web [4], evaluate functional task completion. They measure whether an agent achieves a goal, not whether it experiences the interface the way a human would. This distinction matters for usability research: a system can be functionally correct yet deeply frustrating to human users, and an agent that completes tasks efficiently may still fail to surface the friction points that matter most.

We present **FlowBench**, the first benchmark for measuring behavioral similarity between LLM agents and human users on real-world web usability tasks. FlowBench defines similarity across three dimensions—quantitative usability scores, step-wise difficulty trajectories, and qualitative reasoning—and aggregates them into a single interpretable score. To establish a human baseline, we conduct a structured user study with 20 participants across 10 websites, capturing SUS scores, step-wise SEQ ratings, and Think Aloud transcripts. A persona-matched LLM agent is then evaluated on the same tasks, enabling direct one-to-one comparison at the individual level.

This paper makes the following contributions:

- We introduce **FlowBench**, a benchmark comprising 10 websites and 10 tasks with human baseline data from 20 participants, designed to measure behavioral similarity between LLM agents and human users in web usability contexts.
- We propose a **three-dimensional similarity metric** combining SUS score alignment, SEQ trajectory correlation, and

*All authors contributed equally to this research.
†Work done at UCL Nexus Labs.
‡Corresponding author.

Think Aloud semantic similarity, aggregated into a unified FlowBench score.
- We define a **persona-matched evaluation protocol** enabling one-to-one comparison between individual human participants and corresponding agent instances initialized with matched persona profiles.
- We instantiate the benchmark using **OpenFlo** as the reference agent harness and report preliminary results, establishing a baseline for future agent comparisons.

2 Related Works

2.1 Automated User Experience Evaluation

Traditional UX evaluation methods, such as laboratory studies, provide high-fidelity insights but are resource-intensive. Early automation efforts relied on static analysis tools or clickstream logging, which capture *what* users do but fail to explain *why*. Recent approaches have attempted to leverage AI for simulation, yet many lack the granular, human-like perception required to identify subtle usability friction. Work such as UXAgent by Lu et al. [7] automates UX evaluation using DOM-based agents, while Luera et al. [8] benchmark MLLMs against humans on static screenshot judgments. These works motivate FlowBench: they demonstrate both the promise of automated evaluation and the unresolved question of *how closely* agent behavior aligns with real human users during live interaction.

Table 1: Comparison of web evaluation approaches

Feature	WebArena [15]	UXAgent [7]	FlowBench
Goal	Task completion accuracy	Automated UX reporting	Human-agent similarity
Human Baseline	No	No	Yes (20 participants)
Similarity Metric	None	None	SUS + SEQ + Think Aloud
Perception	DOM-based	DOM-based	Visual (agent-agnostic)
Persona Matching	No	No	Yes (1-to-1)

2.2 Benchmarking LLM Agent Behavior

Benchmarks for web agents have largely focused on functional correctness. WebArena [15] and Mind2Web [4] evaluate whether agents complete tasks on realistic websites, measuring success rates against ground-truth outcomes. While these benchmarks have driven significant progress in agent capability, they do not measure whether agents *experience* interfaces in human-like ways. FlowBench is distinct in measuring the process of interaction—how difficulty is distributed across steps, how reasoning is verbalized—rather than only whether a terminal goal is reached.

2.3 MLLM as a Judge

Luera et al. [8] benchmark MLLMs against humans in evaluations of 30 unique user interfaces using static screenshots, finding that MLLMs perform moderately well on overall system assessments but exhibit systematic weaknesses in judging ease-of-use. This finding directly motivates FlowBench: rather than static screenshot analysis, FlowBench captures dynamic interaction across a full task session, enabling measurement of the precise dimension—ease-of-use judgment—where static approaches fall short.

2.4 System Usability Scale (SUS)

The System Usability Scale (SUS), developed by Brooke [2], is a widely used standardized 10-item questionnaire for measuring the perceived usability of systems. The SUS is well-researched and heavily documented [5], and has proven reliable and effective [1, 3]. FlowBench uses SUS as one of its three similarity dimensions, comparing agent-reported SUS scores against human participant scores on identical tasks.

Table 2: The System Usability Scale [2]

	Statement	1	2	3	4	5
1	I think that I would like to use this system frequently.	○	○	○	○	○
2	I found the system unnecessarily complex.	○	○	○	○	○
3	I thought the system was easy to use.	○	○	○	○	○
4	I think that I would need the support of a technical person to be able to use this system.	○	○	○	○	○
5	I found the various functions in this system were well integrated.	○	○	○	○	○
6	I thought there was too much inconsistency in this system.	○	○	○	○	○
7	I would imagine that most people would learn to use this system very quickly.	○	○	○	○	○
8	I found the system very cumbersome to use.	○	○	○	○	○
9	I felt very confident using the system.	○	○	○	○	○
10	I needed to learn a lot of things before I could get going with this system.	○	○	○	○	○

1: Strongly Disagree, 5: Strongly Agree

The System Usability Score is computed as:

$$\text{SUS} = 2.5 \cdot \left(\sum_{n=1}^{5} (Q_{2n-1} - 1) + \sum_{n=1}^{5} (5 - Q_{2n}) \right) \quad (1)$$

With a dataset of 446 studies and over 5000 individual SUS responses, the Sauro-Lewis Curved Grading Scale [13] provides a reference distribution for interpreting scores.

Table 3: Sauro-Lewis Curved Grading Scale (CGS)

Score	Grade	%	Score	Grade	%
> 84.1	A+	96–100	71.1–72.5	C+	60–64
80.8–84.0	A	90–95	65.0–71.0	C	41–59
78.9–80.7	A-	85–89	62.7–64.9	C-	35–40
77.2–78.8	B+	80–84	51.7–62.6	D	15–34
74.1–77.1	B	70–79	< 51.7	F	0–14
72.6–74.0	B-	65–69			

2.5 Single Ease Question (SEQ)

Table 4: The Single Ease Question (SEQ) [12]

	Statement	1	2	3	4	5	6	7
1	Overall, how difficult or easy was this task?	○	○	○	○	○	○	○

1: Very Difficult, 7: Very Easy

Sauro and Dumas [12] demonstrated that SEQ is the superior choice for high-frequency, iterative UX evaluation contexts, correlating strongly with task completion time ($r = -0.90$) and error rates ($r = -0.84$). A mean score of ≈ 5.5 is considered the threshold for a good experience [11]. FlowBench uses step-wise SEQ as its second similarity dimension, measuring whether agents and humans experience friction at the same points in a task.

3 FlowBench

3.1 Benchmark Design

FlowBench comprises 10 websites spanning e-commerce and SaaS domains, selected to represent diverse interface complexity, interaction patterns, and user task types. For each website, a single website-specific task is defined (e.g., locating a product, checking availability, or configuring an account setting). Tasks are scoped to be completable within a single session without requiring account creation or payment, consistent with ethical constraints on automated evaluation.

3.2 Human Study Protocol

Participants and Personas. We recruit 20 participants who each complete all 10 tasks across the 10 benchmark websites (one task per website). Prior to the study, each participant completes a demographic and technology-use questionnaire, which is used to construct a structured *persona profile* capturing their digital literacy, browsing habits, and familiarity with the task domain. These profiles serve as the basis for constructing matched agent personas (Section 3.3).

Data Collection. Each session is screen-recorded. Participants follow a concurrent Think Aloud protocol [9], verbalizing their intentions, confusions, and observations throughout the task. Sessions are transcribed post-hoc. After each discrete interaction step, participants provide a step-wise SEQ rating on a 1–7 scale. Upon task completion or failure, participants complete the 10-item SUS questionnaire (Table 2).

Inter-Rater Agreement. Because individual usability judgments naturally vary across personas, each participant is treated as an independent rater rather than aggregated into a single ground truth. This preserves within-population variance and enables persona-level comparison with matched agent instances.

3.3 Agent Evaluation Protocol

We evaluate agents against the human baseline using **OpenFlo** as the reference harness. OpenFlo is a GUI-grounded MLLM framework built on Avenir-Web [6] that executes web tasks through visual perception rather than DOM parsing, interacting with real web interfaces through coordinate-based grounding. This enables human-like interaction with pages as they are rendered—including visual layout, styles, and dynamic elements—rather than relying on simplified HTML representations.

For each human participant p_i with persona profile π_i, a corresponding agent instance a_i is initialized with a system prompt encoding π_i. The agent completes the same task on the same website, producing a step-wise SEQ trace, a post-task SUS score, and a Think Aloud transcript generated as concurrent reasoning output. This one-to-one matching enables fine-grained comparison at the individual level.

3.4 FlowBench Similarity Metrics

For each matched human-agent pair (p_i, a_i), we compute similarity across three dimensions and aggregate them into a single score.

SUS Similarity (S_{SUS}). We compute the normalized absolute difference between the human and agent SUS scores:

$$S_{\text{SUS}}(p_i, a_i) = 1 - \frac{|\text{SUS}_{p_i} - \text{SUS}_{a_i}|}{100} \tag{2}$$

A score of 1.0 indicates identical SUS assessments; 0.0 indicates maximum divergence.

SEQ Trajectory Similarity (S_{SEQ}). Step-wise SEQ scores form a sequence over the course of the task. We compute the Pearson correlation between the human and agent SEQ sequences, normalized to $[0, 1]$:

$$S_{\text{SEQ}}(p_i, a_i) = \frac{1 + r(\text{SEQ}_{p_i}, \text{SEQ}_{a_i})}{2} \tag{3}$$

where r denotes Pearson's correlation coefficient. This captures whether the agent experiences friction at the same interaction steps as the human, independent of absolute score magnitude.

Think Aloud Similarity (S_{TA}). Think Aloud transcripts are encoded using a pre-trained sentence embedding model. Document-level representations are obtained by mean-pooling step-wise embeddings. Similarity is computed as the cosine similarity between human and agent transcript embeddings:

$$S_{\text{TA}}(p_i, a_i) = \cos(\mathbf{e}_{p_i}, \mathbf{e}_{a_i}) \tag{4}$$

FlowBench Score. The three dimensions are aggregated using equal weighting:

$$\text{FlowBench}(p_i, a_i) = \frac{S_{\text{SUS}} + S_{\text{SEQ}} + S_{\text{TA}}}{3} \tag{5}$$

A FlowBench score of 1.0 indicates perfect behavioral alignment between the agent and its matched human participant. We report aggregate benchmark performance as the mean FlowBench score across all 20 participant-agent pairs and all 10 websites, providing both per-website and per-dimension breakdowns.

4 Preliminary Results

We report preliminary findings from a pilot evaluation of FlowBench on two websites prior to the full human study. The pilot uses OpenFlo as the agent harness and focuses on characterizing agent-side behavior across the three FlowBench dimensions. Full human baseline comparison will follow upon study completion; the results here establish the format and feasibility of the benchmark.

4.1 Pilot Setup

Two websites were selected for the pilot: **Discogs** (a music marketplace and database) and **Recreation.gov** (a government permit and reservation platform). These represent contrasting usability profiles—one with clear information hierarchy, the other with complex interactive widgets. For each website, the agent was initialized with a general user persona and executed the assigned task using OpenFlo's visual grounding pipeline.

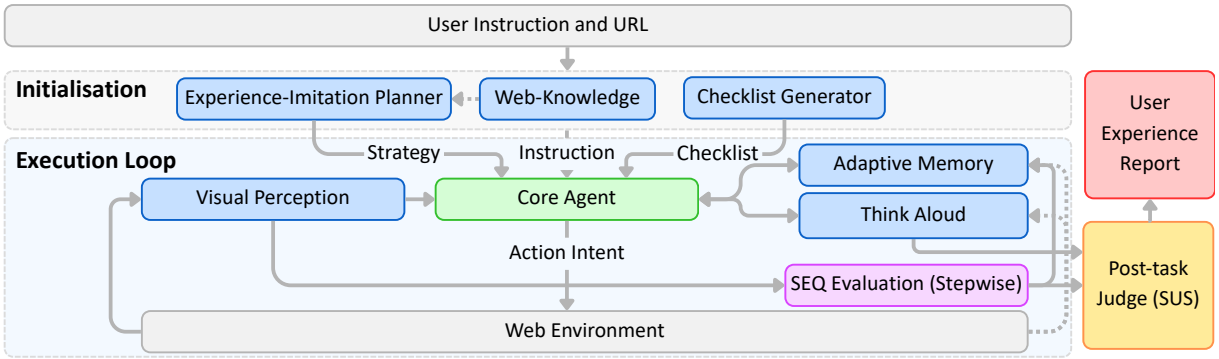

Figure 2: FlowBench evaluation pipeline. Human participants and persona-matched agents complete identical tasks on the same websites. Their SUS scores, SEQ trajectories, and Think Aloud transcripts are compared across three dimensions to compute a unified FlowBench score.

4.2 Agent-Side Results

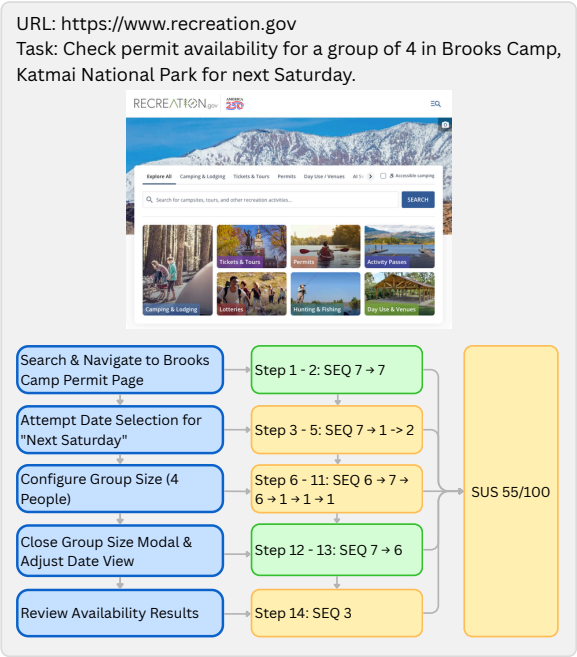

Figure 3: Recreation.gov agent task execution showing bimodal SEQ performance across 14 steps. Green boxes indicate smooth interactions (SEQ ≥ 6.5), yellow boxes indicate friction points (SEQ ≤ 3.3), resulting in a final SUS score of 55.0.

Table 5 summarizes the agent-side usability scores for both pilot websites.

Table 5: Agent-side pilot results (OpenFlo harness)

Website	Avg SEQ	SUS Score	SUS Grade
Discogs	6.0 / 7	87.5	A+
Recreation.gov	4.1 / 7	55.0	D

Discogs. The agent completed the task (locating submission guidelines) in 4 steps with an average SEQ of 6.0/7 and a SUS score of 87.5. The Think Aloud trace indicated a coherent strategy: the agent reasoned correctly that documentation links would appear in the footer rather than the primary navigation, bypassing the commercial marketplace interface. The sole friction point (SEQ = 5 at step 2) arose from a cookie consent modal, which temporarily interrupted the interaction flow.

Recreation.gov. The agent was tasked with checking campsite availability for a group of 4 at Brooks Camp, Katmai National Park. The task exposed a website where visual clarity masks functional defects. Initial navigation succeeded (Steps 1–2: SEQ = 7), but the system degraded during date selection (Steps 3–5: SEQ = 7 ⟶ 1 ⟶ 2) and group size configuration (Steps 6–11: SEQ = 6 ⟶ 7 ⟶ 6 ⟶ 1 ⟶ 1 ⟶ 1). The agent's Think Aloud captured the precise failure mode: "*While the DOM element is clearly visible and correctly identified, the lack of response creates a total block.*" The task concluded with a SUS score of 55.0, corresponding to a grade D.

4.3 FlowBench Score Computation

To illustrate how the FlowBench score will be computed once human baseline data is collected, we present the expected structure for a single participant-agent pair on Recreation.gov. Assume a human participant p_i with a matched persona completes the same task and produces a SUS score SUS_{p_i}, a step-wise SEQ sequence SEQ_{p_i}, and a Think Aloud transcript $\mathbf{e}_{p_i}$. The FlowBench score for this pair would then be:

$$\text{FlowBench}(p_i, a_i) = \frac{S_{\text{SUS}} + S_{\text{SEQ}} + S_{\text{TA}}}{3} \tag{6}$$

The pilot results demonstrate that the agent's SEQ trajectory on Recreation.gov exhibits a distinctive bimodal pattern—high scores in early navigation followed by sharp drops during widget interaction. We hypothesize that human participants with similar task familiarity will exhibit correlated friction patterns at the same steps, yielding high S_{SEQ} values. Conversely, the Discogs task's smooth trajectory may show higher variance against diverse human personas, given that navigation strategies for documentation lookup differ more widely across users.

Full FlowBench scores across all 20 participant-agent pairs will be reported upon completion of the human study.

5 Conclusion

We presented **FlowBench**, a benchmark for measuring behavioral similarity between LLM agents and human users on real-world website tasks. By collecting structured usability data—SUS scores, step-wise SEQ trajectories, and Think Aloud transcripts—from 20 human participants across 10 websites, and comparing them against persona-matched agent instances through a unified three-dimensional similarity metric, FlowBench provides the community with a principled, reproducible framework for evaluating the fidelity of synthetic users in web usability research. Our preliminary results demonstrate the benchmark's feasibility and surface clear contrasts in agent behavior across websites with differing usability profiles, motivating the hypotheses that will be tested in the full study.

5.1 Future Works

Full Benchmark Evaluation. The immediate next step is the completion of the human study and computation of FlowBench scores across all 20 participant-agent pairs. This will establish the first empirical characterization of human-agent behavioral alignment in web usability contexts and validate or challenge the target threshold of 0.7.

Multi-Agent Comparison. While this work uses OpenFlo as the sole reference harness, FlowBench is agent-agnostic. Future work will evaluate multiple agents and underlying LLMs through the same benchmark, enabling direct comparison of which architectures and models best approximate human usability behavior.

Dimension Weighting. Current FlowBench scores use equal weighting across SUS, SEQ, and Think Aloud similarity. Future work will investigate whether differential weighting—informed by the relative predictive validity of each dimension against ground-truth usability outcomes—improves the benchmark's discriminative power.

Persona Diversity. The current study recruits a general participant pool. Future iterations of FlowBench will include participants stratified by digital literacy, accessibility needs, and domain familiarity, enabling measurement of whether agent-human similarity varies systematically across user populations.

Longitudinal Tracking. As websites evolve and models improve, FlowBench can serve as a longitudinal tracking tool, measuring whether improving agent capabilities translate into improved behavioral alignment with human users over time.

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References

[1] Aaron Bangor, Philip T. Kortum, and James T. Miller. 2008. An Empirical Evaluation of the System Usability Scale. *International Journal of Human-Computer Interaction* 24, 6 (2008), 574–594. doi:10.1080/10447310802205776

[2] John Brooke. 1996. SUS: A Quick and Dirty Usability Scale. In *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland (Eds.). Academic Press, London, UK, 189–194.

[3] John Brooke. 2013. SUS: A Retrospective. *Journal of Usability Studies* 8, 2 (2013), 29–40.

[4] Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Samuel Stevens, Boshi Wang, Huan Sun, and Yu Su. 2023. Mind2Web: Towards a Generalist Agent for the Web. arXiv:2306.06070 [cs.CL] https://arxiv.org/abs/2306.06070

[5] James R. Lewis. 2018. The System Usability Scale: Past, Present, and Future. *International Journal of Human-Computer Interaction* 34, 7 (2018), 577–590. doi:10.1080/10447318.2018.1455307

[6] Aiden Yiliu Li, Xinyue Hao, Shilong Liu, and Mengdi Wang. 2026. Avenir-Web: Human-Experience-Imitating Multimodal Web Agents with Mixture of Grounding Experts. arXiv:2602.02468 [cs.AI] https://arxiv.org/abs/2602.02468

[7] Yuxuan Lu, Bingsheng Yao, Hansu Gu, Jing Huang, Jessie Wang, Yang Li, Jiri Gesi, Qi He, Toby Jia-Jun Li, and Dakuo Wang. 2025. UXAgent: A System for Simulating Usability Testing of Web Design with LLM Agents. arXiv:2504.09407 [cs.CL] https://arxiv.org/abs/2504.09407

[8] Reuben A. Luera, Ryan Rossi, Franck Dernoncourt, Samyadeep Basu, Sungchul Kim, Subhojyoti Mukherjee, Puneet Mathur, Ruiyi Zhang, Jihyung Kil, Nedim Lipka, Seunghyun Yoon, Jiuxiang Gu, Zichao Wang, Cindy Xiong Bearfield, and Branislav Kveton. 2025. MLLM as a UI Judge: Benchmarking Multimodal LLMs for Predicting Human Perception of User Interfaces. arXiv:2510.08783 [cs.HC] https://arxiv.org/abs/2510.08783

[9] Jakob Nielsen. 1993. *Usability Engineering*. Academic Press, Boston, MA.

[10] Jeffrey Rubin and Dana Chisnell. 2008. *Handbook of Usability Testing: How to Plan, Design, and Conduct Effective Tests* (2nd ed.). Wiley Publishing, Inc., Indianapolis, IN.

[11] Jeff Sauro. 2012. 10 Things To Know About The Single Ease Question (SEQ). MeasuringU. https://measuringu.com/seq10/ Accessed: February 10, 2026.

[12] Jeff Sauro and Joseph S. Dumas. 2009. Comparison of Three One-Question, Post-Task Usability Questionnaires. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '09)*. ACM, New York, NY, USA, 1599–1608. doi:10.1145/1518701.1518946

[13] Jeff Sauro and James R. Lewis. 2016. *Quantifying the User Experience: Practical Statistics for User Research* (2nd ed.). Morgan Kaufmann, Cambridge, MA, USA.

[14] Dakuo Wang, Ting-Yao Hsu, Yuxuan Lu, Hansu Gu, Limeng Cui, Yaochen Xie, William Headean, Bingsheng Yao, Akash Veeragouni, Jiapeng Liu, Sreyashi Nag, and Jessie Wang. 2025. AgentA/B: Automated and Scalable Web A/BTesting with Interactive LLM Agents. arXiv:2504.09723 [cs.HC] https://arxiv.org/abs/2504.09723

[15] Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024. WebArena: A Realistic Web Environment for Building Autonomous Agents. arXiv:2307.13854 [cs.AI] https://arxiv.org/abs/2307.13854

Appendix: Research Methods & System Prompts

A. System Prompts)

A.1 Step-Wise Evaluation (SEQ & Multi-Metric Prompt)

This prompt is utilized during the Task Execution and Think Aloud phase. It instructs the Avenir-Web Use Agent to evaluate four distinct dimensions of usability after every discrete browser action.

System Prompt

ROLE: You are an Autonomous Browser Use Agent (BUA) equipped with a UX Evaluation layer. Your goal is to execute a defined user flow while meticulously evaluating the user experience across multiple dimensions for every action.

EVALUATION METRICS (1-7 Scale):

- SEQ (Single Ease Question): Overall ease of completing this action.
- Efficiency: Speed and directness of the action (Numeric score + 1-2 sentence qualitative assessment).
- Clarity: How clear and understandable the UI element or feedback was (Numeric score + 1-2 sentence qualitative assessment).
- Confidence: User's certainty about the action and its outcome (Numeric score + 1-2 sentence qualitative assessment).

OPERATIONAL GUIDELINES:

- Thinking Log: Identify specific DOM elements, note latency/layout shifts, and explain how efficiency, clarity, and confidence were affected.
- Friction Indicators: Tag interactions with: waiting, searching, retrying, scrolling, confusion, error, ambiguity, or uncertainty.

A.2 Post-Task Synthesis (SUS Evaluation Prompt)

Following task completion or failure, the UX Research & Data Synthesis Agent uses this prompt to map the micro-metrics gathered in Step A.1 to the 10-item System Usability Scale (SUS).

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System Prompt

ROLE: You are a UX Research & Data Synthesis Agent. Your role is to analyze a completed user flow session and transform micro-metrics (SEQ, Efficiency, Clarity, Confidence) into a macro-level SUS report.

ENHANCED MAPPING LOGIC:

- SEQ $\rightarrow$ Base Assessment: High average SEQ (≥ 5.0) maps to positive SUS scores; Low average SEQ (< 4.0) maps to negative SUS scores.
- Efficiency $\rightarrow$ Item 8 (Cumbersomeness): Low efficiency scores indicate high cumbersomeness.
- Clarity $\rightarrow$ Item 2 (Complexity) & Item 3 (Ease of Use): Low clarity scores trigger high complexity ratings.
- Confidence $\rightarrow$ Item 9 (User Confidence): Directly influences the perception of system reliability and frequency of use.

CRITICAL MANDATE: Base the evaluation on AVERAGE scores across the session. A few outliers should not overshadow an otherwise successful journey.

Avenir-Web Framework

The following sections detail the specific prompts used within the AVENIR-WEB framework to manage strategic reasoning, execution, and state monitoring.[6]

A.3 Core Interaction (System Prompt)

As OpenFlo is built upon the Avenir-Web framework, the agent utilizes this system prompt to define its persona, tool capabilities, and operational rules for navigating live web interfaces.

System Prompt

```
ROLE: You are an autonomous web agent capable of
perceiving web interfaces and executing multi-step
action sequences.

OPERATIONAL RULES:
    • Action Limit: Execute only one action per turn
      with pixel coordinates.
    • Modals: Close or accept blocking modals,
      overlays, or cookie banners first.
    • Coordinates: You MUST provide coordinate [x,y]
      for every CLICK, HOVER, or TYPE action; do not
      rely on text labels alone.
    • Dropdowns: For <select> elements, you MUST use
      the 'select' action directly; do NOT use 'click'
      to open dropdowns.
    • Navigation: Do NOT use GOTO for URL navigation.
    • Termination: When objectives are achieved,
      TERMINATE with status 'success'.

SCREEN SPECIFICATION: 1000x1000, origin (0,0)
top-left.
```

A.4 Checklist Generation Prompt

This prompt is used during the initialization phase to decompose high-level user goals into verifiable atomic milestones, ensuring long-term goal focus.

System Prompt: Task Decomposition Agent

```
ROLE: You are a Task Decomposition Agent. Your goal
is to create 2-6 atomic outcome states based STRICTLY
on the task description.

RULES:
    • Outcome-Based: Each item must be an observable
      goal state, not a description of an action.
    • Conciseness: Maximum 10 words per item; short
      and specific.
    • Fidelity: DO NOT invent requirements not
      explicitly mentioned in the task.
    • Status Formatting: Status must be lowercase:
      pending, in_progress, completed, or failed.
```

A.5 Checklist Update Prompt

This prompt synchronizes the checklist status with the environment state after every interaction step to prevent navigational drift.

GOAL: Update the checklist based on the latest action, its success/failure status, and the observed page state.
UPDATE RULES:

- Mapping Logic: Use *completed* for fully satisfied goals, *in_progress* for partially finished tasks, and *failed* if an action directly leads to an error.
- Precision: Update exactly ONE item per action—the one most directly affected by the preceding operation.

A.6 Experience-Imitation Planning (EIP) Search

The EIP module executes a strategic search to incorporate site-specific procedural knowledge before the execution loop begins, emulating informed human users.

STRATEGY: Perform a targeted search for the website's help documentation, community forums, or user guides to identify site-unique interaction patterns.
NARRATIVE FLOW:

- Exploration: Search official documentation or relevant community-sourced guidance.
- Roadmap Generation: Summarize results into 2–4 actionable, imperative sentences prioritizing visible labels and concrete interaction steps.
- Strategic Injection: Inject this high-level roadmap into the main reasoning context to serve as a strategic anchor for every subsequent action.

A.7 Task Constraints (Safety & Policy)

Standard soft constraints are injected into the User Prompt to ensure ethical interaction and data security during the autonomous evaluation.

POLICY CONSTRAINTS:

- Credentials: Do NOT attempt to log in, sign in, sign up, or provide credentials of any kind.
- Prohibited UI: If a login/sign-in UI is detected (password fields, 'Sign in', 'Log in'), TERMINATE immediately with status 'failure' and reason 'login prohibited'.

B. Detailed Case Study

Task Description. The objective of this automated session was to navigate the Discogs website to locate the specific documentation regarding the submission of new music releases. Unlike a standard retrieval task, this required the agent to bypass the primary commercial interface (marketplace listings and search bars) to identify and access the site's support infrastructure, specifically the "Overview of Submission Guidelines" page.

Session Overview & Infrastructure. The agent initialized on February 9, 2026, utilizing the OpenFlo framework. This system employs a "Three-Engine" architecture designed to mimic human cognitive processes:

- **Reasoning Engine:** *anthropic/claude-sonnet-4.5*. This aligns with the **Experience-Imitation Planning (EIP)** module, which accesses external knowledge to emulate human expert strategies.
- **UX Engine:** *google/gemini-3-flash-preview*. This functions as the **Core Agent** and **UX Researcher**, responsible for reasoning about the UI and calculating usability scores.
- **Checklist Engine:** *qwen/qwen3-vl-8b-instruct*. This powers the **Adaptive Memory and Checklist** component, decomposing high-level goals into atomic milestones.

Phase 1: Strategic Planning. Before executing any browser actions, the agent engaged in Experience-Imitation Planning (EIP). The Reasoning Engine generated a "UX Evaluation Mindmap," correctly

predicting that technical guidelines would likely be located in the footer or a dedicated "Help" section rather than the main navigation.

To track progress, the Checklist Generator established three observable outcome states:

(1) **Homepage Loading**
(2) **Help Section Visibility**
(3) **Submit Guidelines Discovery**

Phase 2: Action-by-Action Breakdown.

Step 1: Handling Interstitials (Cookie Consent).

- **Action:** Click "Accept All" at coordinates (805, 876).
- **Architecture Correlation:** This demonstrates the Visual Perception & Grounding module. Unlike DOM-based agents that might fail due to obfuscated HTML, OpenFlo uses Mixture of Grounding Experts (MoGE) to interact with pixels directly via coordinate-based visual tagging.
- **Status:** The agent adhered to operational rules to close blocking overlays first.

Step 2: Site Infrastructure Scouting.

- **Action:** `scroll_bottom`.
- **Architecture Correlation:** This action was driven by the **Think Aloud** protocol, where the agent reasoned that guidelines are often found in the footer.
- **Result:** The agent bypassed the "visual clutter" of the marketplace to find the structural links.

Step 3: Deep Link Navigation.

- **Action:** Click "Database Guidelines" at (316, 838).
- **Architecture Correlation:** The agent utilized grounded interaction to identify the specific text link that matched the user's intent. This triggered a domain switch to `support.discogs.com`, managed by the agent's Adaptive Memory to maintain context across the transition.

Step 4: Verification and Termination.

- **Action:** `terminate` with status 'success'.
- **Architecture Correlation:** The agent verified the page title "Overview Of Submission Guidelines" against the checklist requirements before terminating, ensuring the objective was achieved.

Final UX Evaluation & Success Metrics. The UX Synthesis Manager processed the interaction data to generate a standardized UX report.

- **Total Steps:** 4.
- **Avg SEQ (Single Ease Question): 6.0 / 7**.
  - This score exceeds the threshold of ≈ 5.5, which is considered a "good" experience.
  - The drop in SEQ (to 5/7) during Step 2 (Scrolling) correlates with the "privacy consent" modal interference, detecting a specific friction point.
- **Final SUS (System Usability Scale): 87.5**.
  - According to the **Sauro-Lewis Curved Grading Scale**, this score translates to a **Grade: A+**.

Complexity and Performance Analysis. The Discogs submission guidelines trajectory serves as a validation of the OpenFlo architecture's ability to handle unstructured exploration tasks. While previous case studies (e.g., Recreation.gov) highlighted challenges with complex widgets, the Discogs task presents a different set of challenges: Information Hierarchy and Visual Noise.

1. Solving Layout Ambiguity via MoGE. The Discogs homepage is dense with marketplace listings and ads. A traditional DOM-based agent might struggle to distinguish between a "Help" link and a commercial product listing. OpenFlo's Visual Perception module (using MoGE) allows the agent to "see" the page layout as a human does, discarding styles and layout ambiguity to locate the footer content accurately.

2. Strategic Navigation via EIP. The task required the agent to ignore the prominent search bar—which is usually the primary interaction point—and instead seek documentation. This behavior is powered by Experience-Imitation Planning (EIP), which allows the agent to emulate the strategy of an informed user who knows that guidelines are typically "infrastructure" links rather than "content" links.

3. State Consistency via Adaptive Memory. Transitioning from the main `www.discogs.com` domain to `support.discogs.com` often resets the DOM context. The Adaptive Memory module ensures the agent retains the original goal ("find submission guidelines") across this boundary, preventing the navigational drift often seen in less capable agents.

By completing this task in a concise 4-step sequence with a high SEQ score, OpenFlo demonstrates that its GUI grounding techniques effectively bridge the gap between functional testing and true user experience evaluation.